

White Paper 8 — Structure-based cross-check by co-folding

Pipeline step: Structure-based cross-check on top hits **Inputs:** top docking hits, lead domain sequences **Outputs:** `psoriasis_cofolding_package.tar.gz` (Boltz-2 YAMLs, scripts, domain FASTA)

1. Purpose

Docking scores carry known biases (rigid receptor, empirical scoring). An independent method that does not share those biases provides a valuable cross-check. This step packages a **co-folding + affinity** validation of the top docking hits using Boltz-2 — a model that predicts the protein–ligand complex structure *and* a binding-affinity estimate from sequence and chemistry alone.

2. The method: Boltz-2 co-folding

Boltz-2 is an open-weights biomolecular structure predictor in the AlphaFold3 family: given protein sequence(s) and ligand(s), it jointly predicts the 3D complex and reports per-interface confidence, together with an optional small-molecule **binding-affinity** head [1]. It builds on the Boltz-1 architecture [2], which democratized AlphaFold3-style complex modeling in an openly-licensed implementation. The affinity head is what makes Boltz-2 a genuine orthogonal check: it scores binding through a learned model rather than a docking scoring function.

3. Why co-folding is a true cross-check

Docking places a ligand into a fixed pocket and scores the pose; co-folding *predicts the entire complex* from sequence, allowing the protein to adopt a ligand-appropriate conformation. Agreement between a docking score and a co-folding affinity/interface-confidence is therefore strong evidence for a genuine binder, while disagreement flags a hit for manual inspection or MD. The approach descends from the AlphaFold2 breakthrough in learned structure prediction [3].

4. Package design

The co-folding package (`psoriasis_cofolding_package.tar.gz`) contains:

- **11 ready RORyt inputs** — YAML specifications for the top-10 docking hits plus the crystal ligand VYI as a positive control, each with the affinity head enabled.
- **Domain constructs** — the exact sequence used for each lead (RORyt LBD, STAT3 core+SH2, JAK3 kinase), matching the docking receptors, provided as FASTA. Using the domain rather than the full-length protein keeps the prediction focused and GPU-efficient.
- **STAT3 and JAK3 templates** — to be populated with their docking hits from the full screen.
- **GPU SLURM array** — one prediction per task; the multiple-sequence-alignment step uses the ColabFold MSA server [4] (or precomputed alignments on an air-gapped node).
- **Cross-check aggregation** — a script that merges Boltz-2 interface confidence (ipTM) and affinity probability with the Vina scores into a single comparison table.

5. Reading the output

Per prediction, three numbers matter: **ipTM** (interface confidence; > 0.5 is the community pass line), **affinity probability** (0–1 binder score, the value to rank by), and **predicted affinity** (log10 IC50 in μM). A hit is corroborated when a strong Vina score coincides with high Boltz-2 binder probability and interface confidence; the VYI positive control should score as a confident binder.

6. Deliverables

- `psoriasis_cofolding_package.tar.gz` — YAML inputs, domain FASTA, GPU SLURM runner, and the docking cross-check aggregation script.

7. Caveats

- The Boltz-2 affinity head caps ligands at 128 heavy atoms; larger ligands (e.g. STAT3 degrader chemistry) yield a structure but no affinity.

- Co-folding confidence is a model estimate, not an experimental measurement; it strengthens or weakens a docking hypothesis but does not confirm binding.

References

1. Passaro et al. Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction. 2025. doi:[10.1101/2025.06.14.659707](https://doi.org/10.1101/2025.06.14.659707)
2. Wohlwend et al. Boltz-1 Democratizing Biomolecular Interaction Modeling. 2024. doi:[10.1101/2024.11.19.624167](https://doi.org/10.1101/2024.11.19.624167)
3. Jumper et al. Highly accurate protein structure prediction with AlphaFold. *Nature*. 2021. doi:[10.1038/s41586-021-03819-2](https://doi.org/10.1038/s41586-021-03819-2)
4. Mirdita et al. ColabFold: making protein folding accessible to all. *Nature Methods*. 2022. doi:[10.1038/s41592-022-01488-1](https://doi.org/10.1038/s41592-022-01488-1)